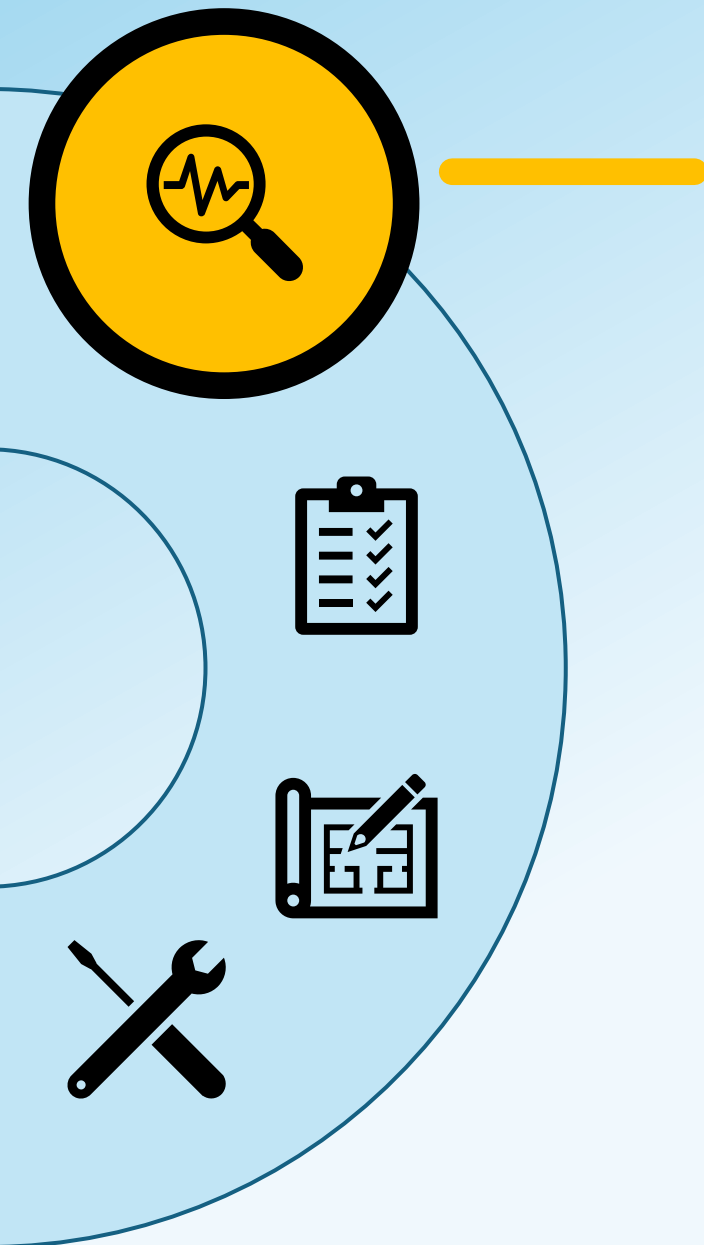


# **REAL-TIME LAN-BASED QUEUING MANAGEMENT & STATISTICAL REPORTING SYSTEM**

**FOR THE BPL-LIEO OF LGU ROSARIO, LA UNION**

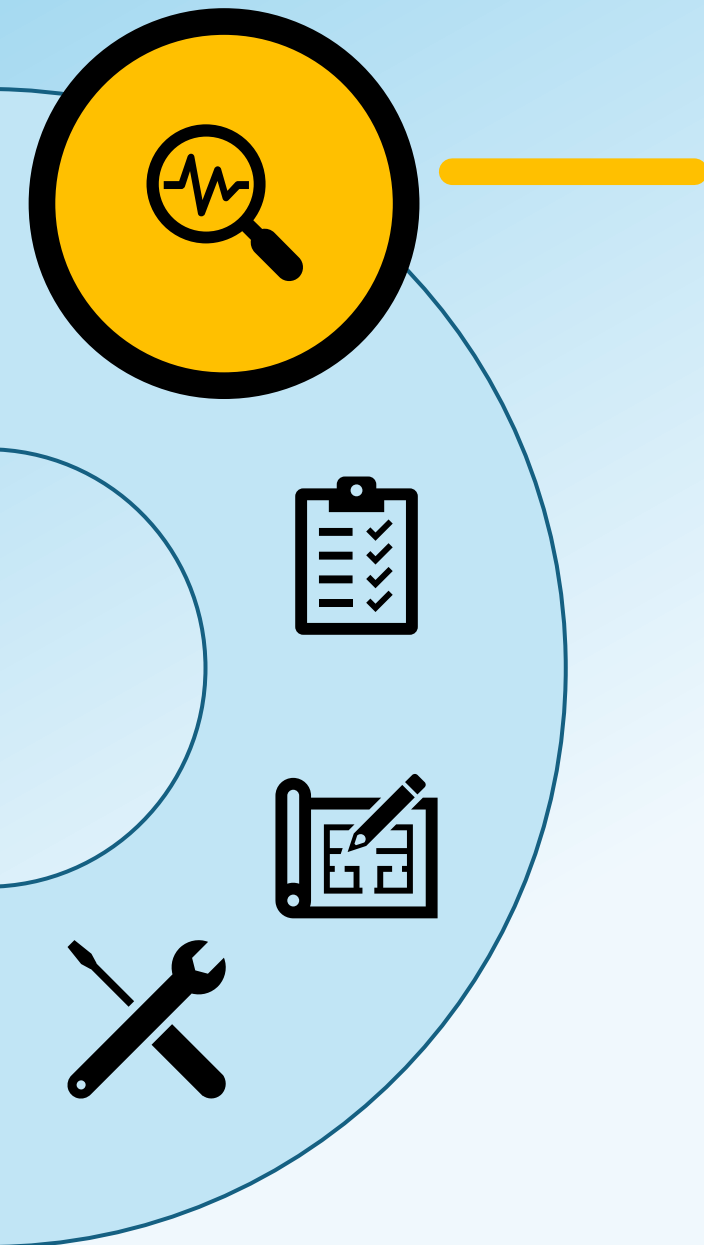
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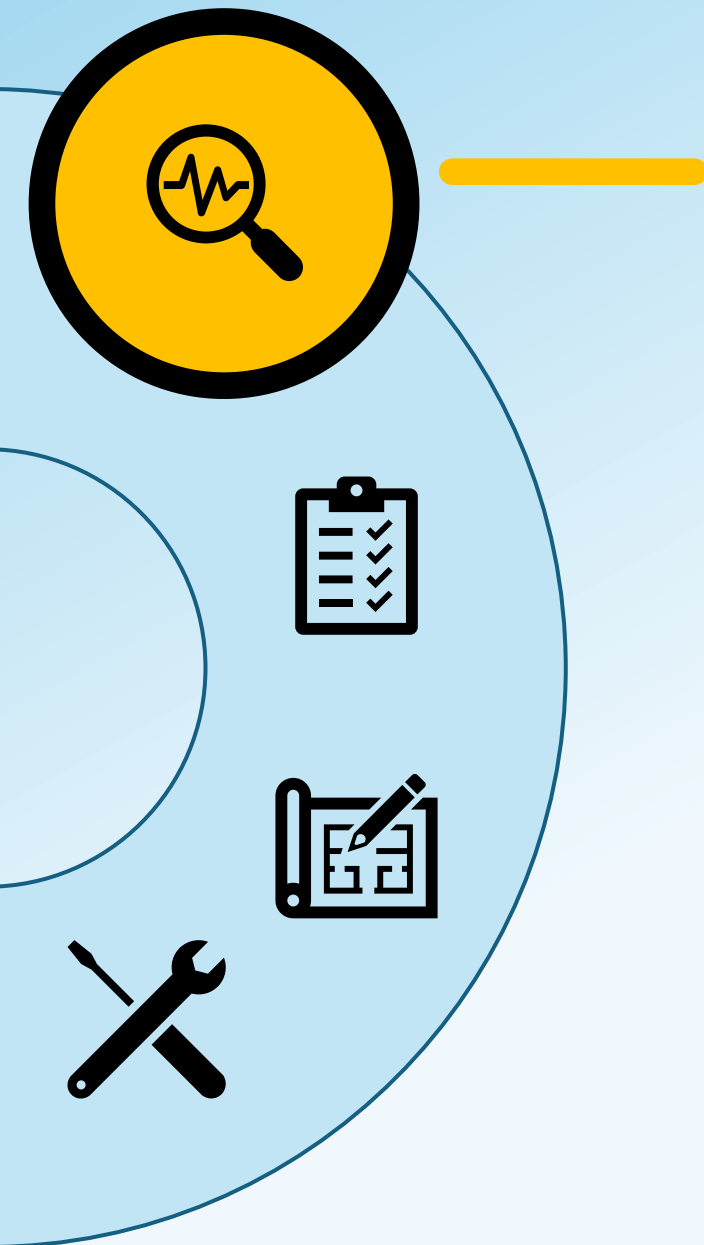
## Situation Analysis: The Current State

- **The Context:** BPL-LIEOs accommodate a massive influx of clients during peak business registration and renewal periods.
- **The Current Method:** Heavy reliance on manual queuing and verbal calling.
- **The Bottlenecks:**
  - Disorganized client flow and overcrowding.
  - Client uncertainty regarding queue position.
  - Staff difficulty in simultaneously processing documents, managing the line, and tracking skipped or returning clients.



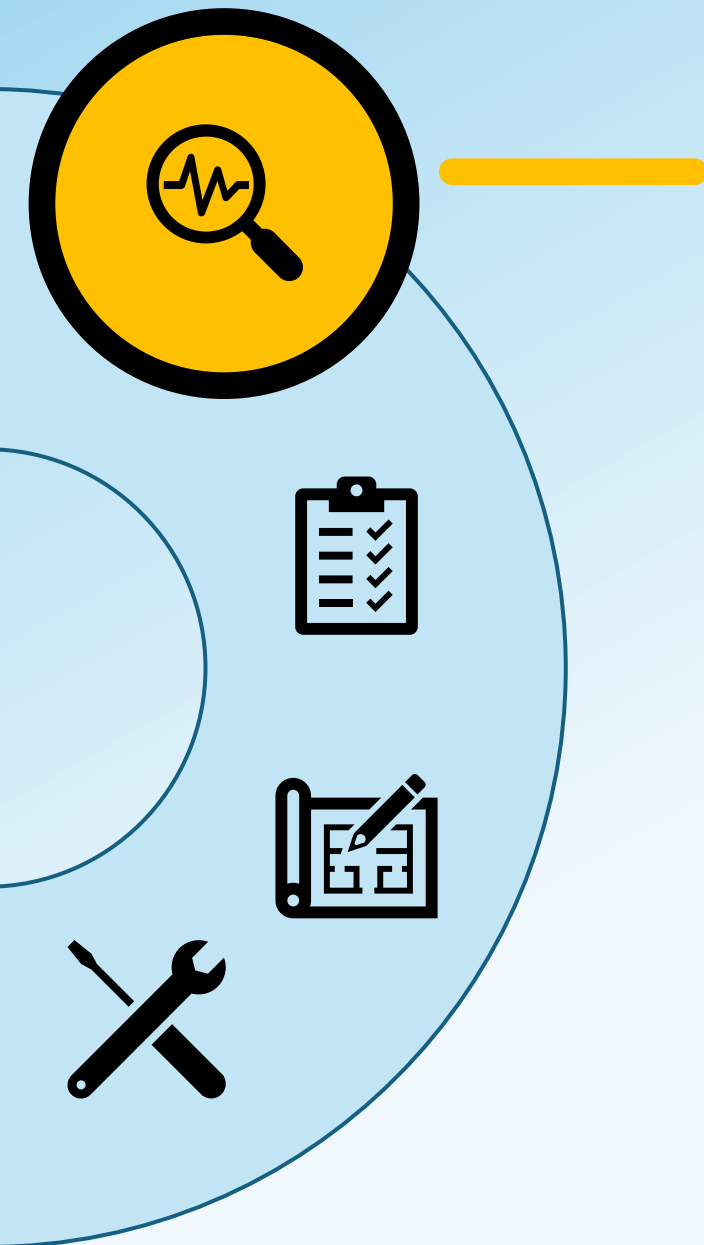
## Situation Analysis: The Research Gap

- **The Assumption:** Many assume any generic, off-the-shelf queuing system (like in banks or fast food) can work for a government office.
- **The Reality:** Generic systems lack the flexibility to handle the unique, multi-stage workflow of the Rosario BPL-LIEO.
- **Specific Missing Features in Generic Systems:**
  - They don't differentiate between New vs. Renewal workflows at the point of ticket generation.
  - They lack a structured mechanism for handling clients with incomplete requirements who must return the next day.
  - There is no localized, real-time tracking of complex transaction statuses.



## Situation Analysis: Our Edge

- **Workflow-Specific Ticket Generation:** Immediately categorizes clients (New vs. Renewal) ensuring they are routed to the correct service process.
- **"Return Tomorrow" & Priority Handling:** A built-in feature specifically designed for the LGU reality. Clients lacking documents can return the next day and be automatically prioritized by staff without creating a new bottleneck.
- **LAN-Based Reliability:**
- **Data-Driven Governance:**



## Situation Analysis: Our Edge

- **Workflow-Specific Ticket Generation:**
- **"Return Tomorrow" & Priority Handling:**
- **LAN-Based Reliability:** Unlike cloud-based web apps, our system operates entirely on a Local Area Network (LAN). It guarantees 100% uptime even during external internet outages, a critical requirement for continuous LGU operations.
- **Data-Driven Governance:** It doesn't just manage lines; the Statistical Reporting Module translates raw queue data into actionable insights (peak hours, staff performance) for LGU administrators.



## Statement of Objectives

- **General Objective:** To design and implement a Real-Time LAN-Based Queuing Management and Statistical Reporting System specifically adapted for the Rosario BPL-LIEO.
- **Specific Objectives:**
  1. To design and develop an automated LAN-based system (Ticket Generator, Staff Dashboard, Public Monitor, Admin Analytics).
  2. To test and improve system performance via alpha and beta testing.
  3. To assess software quality based on ISO/IEC 25010 Standards.



## Research Design

- **Developmental Research:** Systematically designing, building, and refining the queuing software to directly solve the operational bottlenecks of the BPL-LIEO.
- **Descriptive Research:** Utilized to objectively evaluate the system's quality and gather quantitative feedback from both staff and citizens compared to their experience with the manual method.

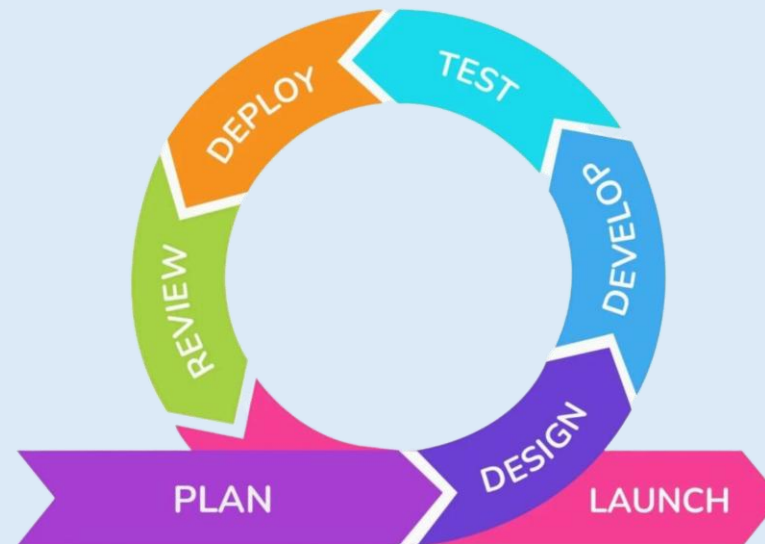


## Materials and Procedure: Tools & Tech Stack

- **Hardware Environment:** Local Area Network (LAN) setup using standard Laptops, Ticket Printers, and Smart TVs.
- **Software Stack:**
  1. Frontend: React.js, HTML5, Vanilla CSS
  2. Backend: Node.js with Express.js
  3. Database: Prisma ORM with SQLite
  4. Real-time Communication: Socket.io

## Materials and Procedure: SDLC

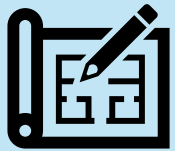
- **Approach:** Agile Software Development Life Cycle (SDLC).
- **Why Agile?** It allows us to build the system incrementally in sprints, ensuring we can easily adapt the software based on direct feedback from BPL-LIEO personnel during development.
- **Phases:** Plan, Design, Develop, Test, Deploy and Review.



# Materials and Procedure: System Design

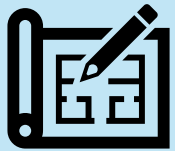
## ➤ Visualizing the Solution:

1. BPMN Diagram: Maps the step-by-step physical and digital interactions from the citizen's arrival to departure.
2. Entity-Relationship Diagrams (ERD): Ensuring robust transaction data storage.
3. Data Flow & Use-Case Diagrams: Defining clear roles for Citizens vs. Staff.



## Materials and Procedure: Testing Procedures

- **Continuous Testing:** Ensuring code quality and real-time Socket.io stability.
- **Alpha Testing:** Conducted in a controlled lab environment by developers to iron out initial logic and UI bugs.
- **Beta Testing:** The crucial phase, deployed on the actual BPL-LIEO LAN environment and tested by actual personnel to validate network performance and usability under real-world conditions.





## Data Gathered

- **Assessment Tool:** A customized survey questionnaire based on the ISO/IEC 25010 Software Quality Model (adapted from Malaya et al., 2022).
- **Respondents (Total N = 60):**
  - **BPL-LIEO Staff:** 10 respondents (Total Enumeration) – evaluating operational efficiency and staff controls.
  - **Citizens / Clients:** 50 respondents (Convenience Sampling) – evaluating public monitor visibility and process transparency.



## Data Analysis

- **Statistical Tool:** Weighted Mean
- **Evaluation Scale:** 5-point Likert Scale assessing seven ISO/IEC 25010 characteristics (Functionality, Usability, Efficiency, Reliability, Security, Maintainability, Portability).
- **Goal:** To determine the descriptive equivalent rating (from Poor to Excellent) and definitively prove the system's acceptability and superiority over the manual method.